

Name - Saikat Sheet
Batch - 25SUB5936_Linux System Programming
ID - 68500

Task 15:

User-Space Program to Test the Custom System Call

Write a C program in user space that tests your new custom system call.

- The program should prompt the user to enter a string.
- It should then use the custom system call to reverse the string.
- Finally, the program should print the reversed string returned by the system call.
- Include in your program:
- Proper error handling to ensure that the system call is invoked correctly.
- Memory management to allocate and free the necessary space for the strings.

Process:

-> In

(\$root@bd310440ba77:/kernel/linux-6.6/kernel/custom_syscall
s) we are adding code hello_syscall.c.

Code:

```
SYSCALL_DEFINE3(reverse_string, const char __user *,
user_input, char __user *, user_output, size_t, len)
{
    char kbuf[256];
    int i;

    if(len == 0 || len >= sizeof(kbuf))
        return -EINVAL;

    /* copy from user -> kernel */
    if(copy_from_user(kbuf, user_input, len))
        return -EFAULT;

    kbuf[len] = '\0';

    printk(KERN INFO "reverse_string called: input=%s
len=%zu\n", kbuf, len);

    /* Reversing the string logic */

    for(i=0; i<len/2; i++)
    {
        char temp = kbuf[i];
        kbuf[i] = kbuf[len - i - 1];
        kbuf[len - i - 1] = temp;
    }

    /* copy kernel -> user */

    if(copy_to_user(user_output, kbuf, len+1))
        return -EFAULT;
```

```
    return 0;
}
```

-> then in (\$root@bd310440ba77:/kernel/linux-6.6# vi arch//x-86/entry/syscalls/syscall_64.tbl) we have to make the entry in the system table

```
455 common reverse_string sys_reverse_string
```

-> then we have to do asm linkage \$vi include/linux/syscalls.h

```
Asmlinkage long sys_reverse_string(const __user, *input, char
__user *output, int len);
```

-> make defconfig

-> make -j\$(nproc)

-> making (\$ vi reverse_str.c)

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/syscall.h>
#include <string.h>
#include <errno.h>

#define __NR_hello 454
#define __NR_reverse_string 455

int main()
{
    long ret;
```

```

syscall(__NR_hello);

char input[256];
char output[256];

printf("\nEnter string to be reversed: ");
fgets(input, sizeof(input), stdin);

input[strlen(input) - 1] = '\0';

size_t len = strlen(input);

ret = syscall(__NR_reverse_string, input, output, len);

if(ret < 0){
    printf("\nError: unable to reverse string\n");
    return 1;
}

printf("\nOriginal string: %s", input);
printf("\nReversed string: %s\n", output);

return 0;
}

```

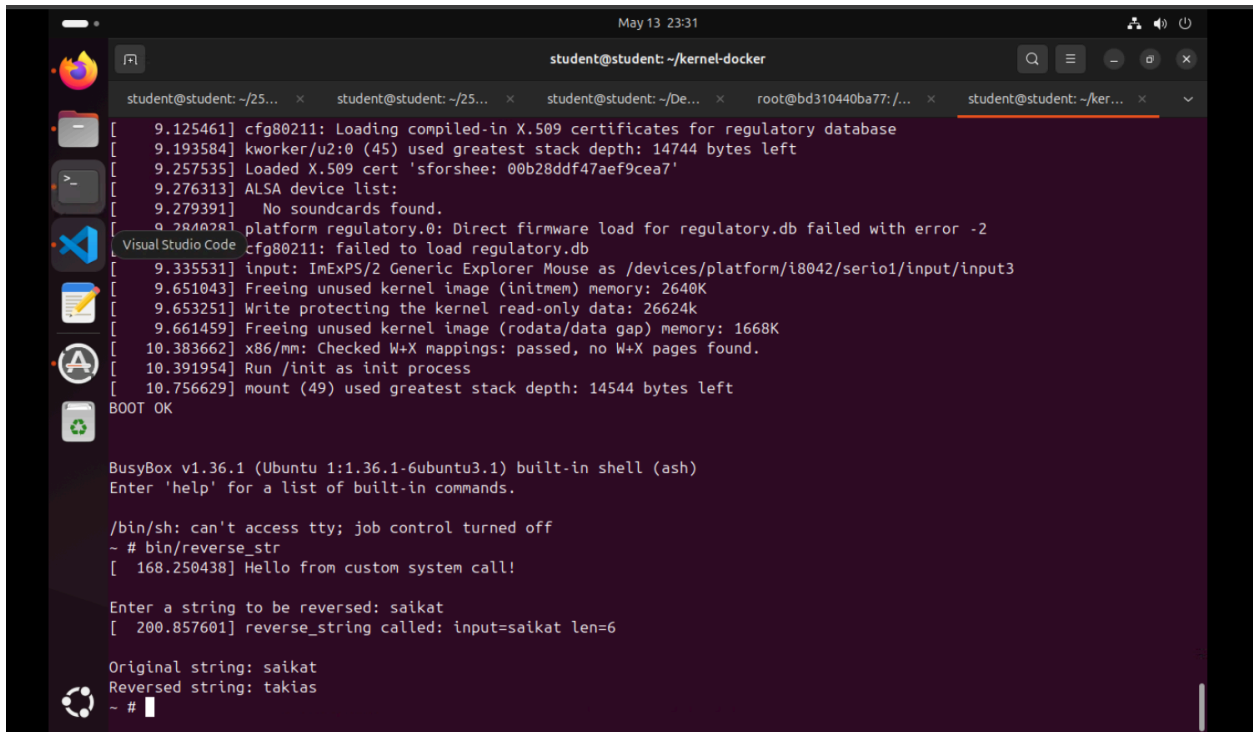
-> gcc -static -o reverse_str reverse_str.c

-> mv reverse_str rootfs/bin/

-> kernel-docker/rootfs\$ find . | cpio -H newC -ov >
 ../initramfs.cpio

```
-> kernel-docker$ qemu-system-x86 -m 512M -kernel  
linux-6.6/arch/x86/boot/bzImage -initrd initramfs.cpio -append  
"console=ttyS0 init=/init" -nographic
```

Output:



```
May 13 23:31
student@student: ~/kernel-docker

[ 9.125461] cfg80211: Loading compiled-in X.509 certificates for regulatory database
[ 9.193584] kworker/u2:0 (45) used greatest stack depth: 14744 bytes left
[ 9.257535] Loaded X.509 cert 'sforshee: 00b28ddf47aef9cea7'
[ 9.276313] ALSA device list:
[ 9.279391]   No soundcards found.
[ 9.284281] platform regulatory.0: Direct firmware load for regulatory.db failed with error -2
[ 9.284281] cfg80211: failed to load regulatory.db
[ 9.335531] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/i8042/serio1/input/input3
[ 9.651043] Freeing unused kernel image (initmem) memory: 2640K
[ 9.653251] Write protecting the kernel read-only data: 26624k
[ 9.661459] Freeing unused kernel image (rodata/data gap) memory: 1668K
[ 10.383662] x86/mm: Checked W+X mappings: passed, no W+X pages found.
[ 10.391954] Run /init as init process
[ 10.756629] mount (49) used greatest stack depth: 14544 bytes left
BOOT OK

BusyBox v1.36.1 (Ubuntu 1:1.36.1-6ubuntu3.1) built-in shell (ash)
Enter 'help' for a list of built-in commands.

/bin/sh: can't access tty; job control turned off
- # bin/reverse_str
[ 168.250438] Hello from custom system call!

Enter a string to be reversed: saikat
[ 200.857601] reverse_string called: input=saikat len=6

Original string: saikat
Reversed string: takias
- #
```